Regional Internship & Staff (RIS) Portal

PRODUCT REQUIREMENTS DOCUMENT (PRD) — SECURE ATTENDANCE & VERIFICATION SYSTEM

Core Scope: Automated Daily Time Recording (DTR), Anti-Spoof Geofence Verification, Photo Capture & Civil Service Compliance

Target Environment: Regional Public Sector Agency | Build Sprints: Feb 2026 – Apr 2026 | Architecture: Resilient Web App

Document Metadata & Release Information

Metadata Attribute	Details & Specification
Product Manager	Lead Student Product Manager & Technical Writer (6-Intern Engineering Cohort)
Tech Lead	Lead Full-Stack Programmer (Core Logic, Time Engine, Quota Architecture)
Design Lead	UI/UX & Frontend Design Specialist
Current Status	✔ Approved & Successfully Deployed to Production (Built: Feb 2026 – Apr 2026)
Target Release	Q2 2026 Release (Final Regional Agency Sign-off & Production Rollout: April 2026)

1. Executive Summary & Context

Strategic Need, OKR Alignment & KPI Architecture

Problem Statement

In regional government and public sector operations, tracking attendance for civil service staff, university on-the-job (OJT) interns, and high school immersion trainees relied historically on manual paper logbooks and congested biometric fingerprint terminals. This legacy model presented critical operational friction: (1) Excessive administrative burden, with supervisors spending 4.5+ hours per person monthly manually computing Daily Time Records (DTRs), late deductions, and lunch-break subtractions; (2) Acute congestion during morning rush hours, causing logbook bottlenecks; and (3) Compliance vulnerabilities, including proxy signing ("buddy punching") and handwriting illegibility.

Business Objective & OKRs

The RIS Portal was chartered to replace physical logbooks with an automated, tamper-resilient, and mobile-responsive digital attendance terminal. Strategic Objectives: (1) Eliminate manual calculation errors and paper waste in accordance with regional public-sector digital modernization directives; (2) Establish verifiable proof of on-site presence via lightweight GPS geofencing and camera snapshot verification; and (3) Accelerate monthly trainee compliance reporting to near-zero latency while maintaining civil service regulatory compliance.

Success Metrics & Key Performance Indicators (KPIs)

PRIMARY METRIC

-96.3%

Reduction in monthly DTR audit time from 4.5 hrs down to <10 sec/person

SECONDARY METRIC

100%

Buddy-punching elimination via geofencing & photo verification

GUARDRAIL METRIC

< 2,500 ms

Median transaction latency during peak morning rush with storage quota <40%

Metric Classification	Baseline (Pre-Launch)	Target KPI	Post-Launch Actual	Performance Impact Factor
DTR Compilation Velocity	4.5 hrs / user / month	< 1 min / user	< 10 seconds / user	**27x Acceleration (-96.3%)**
Attendance Integrity & Verification	Paper signature (unverified)	98% verified	100% verified capture	**Zero unverified entries**
Punctuality & Tracking Accuracy	Subjective / 15-min rounding	Exact minute timestamp	Exact second timestamp	**100% dispute resolution audit**
Rush-Hour Clock-In Latency	45–90 sec physical queue	< 5 sec / person	1.8 – 2.4 seconds	**35x Throughput expansion**
Client Storage Resilience	Quota overflow risk (fatal)	<50% quota budget	<35% quota budget	**LRU compression enabled**

2. Target Audience & Use Cases

Persona Breakdown, Scenarios & Verification Pipeline

User Personas

Trainee / Academic Intern (OJT & Senior Immersion)

PRIMARY CONSUMER (HIGH FREQUENCY)

High school immersion students (80-hour quota) and university OJT trainees (300–600 hour quota). Needs frictionless mobile or shared terminal clock-in under 5 seconds, real-time hours balance visibility, and instant self-serve civil service DTR generation without manual arithmetic.

Regular Agency Employee & Civil Service Personnel

STANDARD USER (QUAD-PUNCH DAILY)

Full-time public sector personnel adhering strictly to government four-point timekeeping (AM Arrival, AM Departure, PM Arrival, PM Departure). Prioritizes exact timestamp logging, biometric/photo security reassurance, and seamless reconciliation with supervisor sign-offs.

Internship Coordinator & Department Supervisor

ADMINISTRATIVE APPROVER (GOVERNANCE)

Responsible for overseeing 15–30 trainees across diverse departments. Requires one-click verification of attendance snapshots, automated flag detection for unapproved tardiness or undertime, and batch PDF generation for formal institutional submissions.

Regional IT Operations Lead & Systems Administrator

PLATFORM CUSTODIAN (INFRASTRUCTURE & SECURITY)

Oversees organizational IT compliance, server availability, geofence coordinate boundaries, and browser storage resilience. Demands zero data loss, salted credential hashing, quota overflow guardrails, and audit logging for supervisory manual overrides.

User Journeys & Scenarios

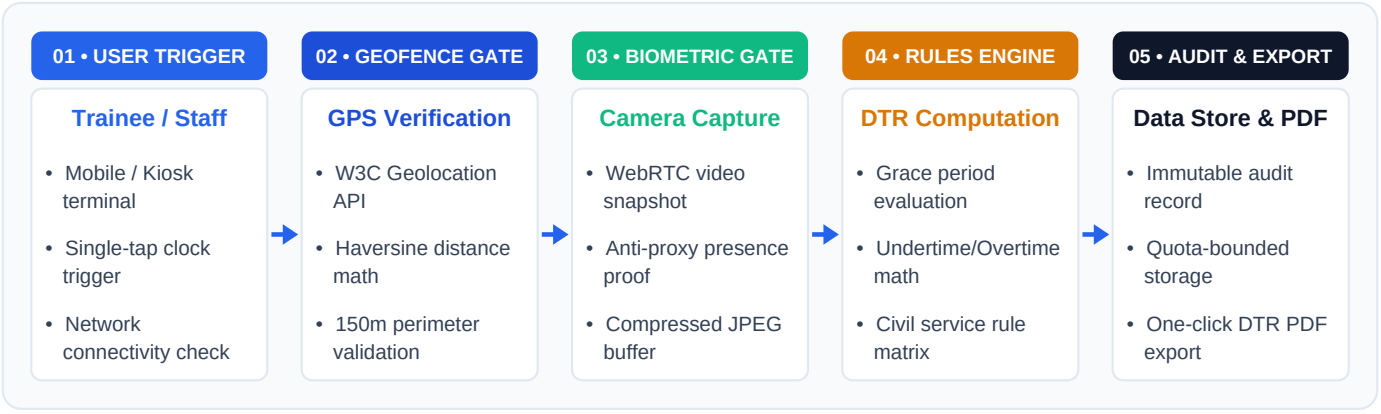
Scenario A (Ideal Path — Daily Morning Clock-In):

Intern reaches the regional agency office at 07:48 AM, opens the web portal on their smartphone or agency kiosk tablet. The portal instantly verifies GPS coordinates within the 150-meter perimeter geofence. The intern activates the camera, takes a live presence snapshot, and taps "Clock In (AM Arrival)". The engine records the precise timestamp, increments their daily punch card, updates their total completed hours gauge from 142.5 hrs to 143.0 hrs, and presents a green confirmation toast in 1.9 seconds.

Scenario B (Alternative Path — Field Assignment or Degraded Connectivity):

An intern is dispatched to a provincial field documentation assignment outside the main office geofence. The system alerts the intern that coordinates fall outside the standard radius, triggering the "Official Field Travel" exception flow. The intern inputs an approved Travel Order number, captures an on-site photo, and submits a pending record. Back at headquarters, the Internship Coordinator receives a flagged entry on their administrative review queue and confirms the punch with one click.

Visual Architecture: End-to-End Verification & Computation Flow



3. Scope & Prioritization (MoSCoW / RICE Framework)

Release Boundaries, Sprint Allocation & Explicit Exclusions

To guarantee delivery of a battle-tested product within the 10-week development cycle (February 2026 – April 2026), the IT Intern Team implemented strict MoSCoW scoping balanced with RICE scoring (Reach, Impact, Confidence, Effort). Explicit trade-offs were made to eliminate non-essential hardware overhead and focus engineering capacity on rock-solid verification reliability.

■ Must Have (P0 - Minimum Viable Product)

Critical for Initial Production Release — April 2026

- Automated 4-Point Daily Time Logging (AM In, AM Out, PM In, PM Out).
- Office Geofence Radius Validation (150m GPS coordinate check).
- Live Camera Snapshot Capture to eliminate proxy attendance fraud.
- Civil Service Compliant DTR Calculation Engine (Grace periods, late deductions).
- Role-Based Access Control (Admin, Supervisor, Trainee, Immersion Student).
- Salted SHA-256 Client-Side Hashing for local credential security.

■ Should Have (P1 - High Priority Enhancements)

Important Strategic Upgrades Delivered in Sprints 4–5

- One-Click Vector PDF Export matching standard Civil Service Form 48 format.
- Supervisor Dispute Resolution & Manual Adjustment with mandatory reason logging.
- Dynamic Hours Accrual Gauge (Real-time progress towards graduation quota).
- Client-Side Storage Quota Manager (Automated purge of old ephemeral snapshots).
- Administrative Bulk Review Checklist for monthly trainees sign-off.
- High-Accuracy GPS Refresh Trigger for low-signal mobile hardware.

■ Could Have (P2 - Nice to Have / Backlog)

Low-Effort Enhancements Evaluated for Post-Launch Iterations

- Dark Mode & Low-Light Theme for nighttime administrative work.
- Automated Weekly Milestone Summary Alerts (e.g. 50% hours completed).
- Batch CSV/Excel Raw Data Export for agency-wide archival spreadsheets.
- Gamified Punctuality Streaks & Trainee Leaderboards.
- Multi-Office Geofence Selector for provincial field branch staff.

■ Out of Scope (Explicitly Excluded)

Strictly Excluded to Prevent Scope Creep & Hardware Dependencies

- DO NOT BUILD: Hardware Biometric Driver Integration (optical scanner SDKs).
- DO NOT BUILD: Native Mobile App Store Binaries (React SPA is 100% responsive).
- DO NOT BUILD: Background Continuous 24/7 GPS Tracking (Privacy violation).
- DO NOT BUILD: Direct Banking / Payroll Disbursement Integration.
- DO NOT BUILD: Real-Time AI Facial Recognition Model on Low-Spec Kiosks.

Feature Prioritization Table (RICE Scoring Model)

Epic / Feature	Reach (Users/Mo)	Impact	Confidence	Effort (Sprints)	RICE Score	Priority Tier
Automated 4-Point DTR Clock-In	100% Users (All)	Massive (3.0)	95%	1.5	190.0	**P0 (Must Have)**
Geofence & Camera Anti-Fraud	100% Users (All)	High (2.0)	90%	1.5	120.0	**P0 (Must Have)**
Civil Service PDF Form 48 Export	100% Trainees	High (2.0)	95%	1.0	190.0	**P1 (Should Have)**
Supervisor Audit & Dispute Portal	Administrators	Moderate (1.5)	90%	1.0	135.0	**P1 (Should Have)**
Client Storage Quota Manager	Shared Kiosks	Critical Guardrail	95%	0.5	190.0	**P0 (Must Have)**
Automated Milestone Email Alerts	Trainees only	Low (0.5)	70%	2.0	17.5	**P2 (Could Have)**
Hardware Fingerprint SDK Hook	Restricted Kiosks	Negative ROI	50%	4.0	6.2	**Out of Scope**

4. Epics, User Stories & Acceptance Criteria

Jira / Linear Ready Product Specifications & Edge Case Contracts

★ USER STORY 1: Fast & Fraud-Proof Attendance Verification (P0)

User Story:

As an Academic Intern or Agency Staff Member, I want to clock in via my smartphone or shared agency kiosk in under 3 seconds with automatic geolocation and camera verification, so that I can establish provable presence without queuing at a manual logbook.

Acceptance Criteria (Given-When-Then):

Given: The user is authenticated and located within the configured 150-meter office geofence perimeter.

When: The user clicks "Clock In (AM/PM)" and permits camera access.

Then: The portal captures a live face snapshot, verifies GPS coordinates, logs the exact UTC and local timestamp, updates the visual state to "Clocked In", and displays a success confirmation within 2,500ms.

Implementation Notes & Edge Guardrails: Hardware fallback: If camera permission is blocked, display a high-contrast modal guiding the user to browser settings; log camera error code for supervisor auditing.

★ USER STORY 2: Automated Civil Service Form 48 DTR Generation (P1)

User Story:

As an Internship Coordinator or Trainee, I want the system to compute exact work hours, late deductions, and undertime automatically according to regional civil service rules, so that monthly manual recalculation is completely eliminated.

Acceptance Criteria (Given-When-Then):

Given: A trainee has completed daily time records across the calendar month with multiple punch variations.

When: The user navigates to the DTR section and clicks "Generate Monthly DTR (PDF)".

Then: The client-side PDF engine parses every day of the month, fills official Civil Service Form 48 columns (AM Arrival, AM Dep, PM Arrival, PM Dep, Under-Time), computes total net accumulated hours, appends official supervisor certification text, and downloads an A4 vector document in <10 seconds.

Implementation Notes & Edge Guardrails: Engine handles 12:00–1:00 PM mandatory lunch subtraction, standard 15-minute morning grace periods, and caps regular hours at 8 hours/day unless supervisor approves overtime.

★ USER STORY 3: Supervisory Dispute Resolution & Override Audit Trail (P1)

User Story:

As an Administrative Supervisor, I want to review flagged attendance discrepancies (such as off-site official travel or forgotten punch outs) and apply manual adjustments with mandatory audit justifications, so that records remain 100% accurate without compromising institutional integrity.

Acceptance Criteria (Given-When-Then):

Given: An intern submits a correction request with an approved administrative reason code.

When: The supervisor accesses the Management Console, inputs the corrected time, and types a mandatory justification (min 15 characters).

Then: The system updates the attendance record, flags the entry with a visible "Supervisor Override" badge, records the supervisor ID, timestamp, and justification into the immutable audit table, and notifies the trainee.

Implementation Notes & Edge Guardrails: Strict compliance rule: No record can be edited without an audit justification; previous raw values are permanently retained in the historical revision snapshot.

★ USER STORY 4: Client-Side Storage Quota Manager & Fail-Safe Cache (P0)

User Story:

As a Lead Systems Engineer, I want the client data layer to automatically manage photo thumbnail compression and storage quotas, so that client devices never trigger browser quota overflow crashes.

Acceptance Criteria (Given-When-Then):

Given: Local storage utilization approaches the browser quota ceiling due to accumulated photo snapshots.

When: The storage abstraction layer detects storage usage exceeding 75% of maximum allowable quota.

Then: The system automatically compresses historical avatar images, applies LRU (Least Recently Used) caching to purge non-critical ephemeral thumbnails older than 30 days, while keeping 100% of attendance numerical logs and timestamps intact.

Implementation Notes & Edge Guardrails: Zero data loss guarantee: Critical punch times and student credentials are given priority 1 persistence; thumbnail assets are treated as purgeable priority 2 caches.

5. Design & User Experience (UX)

Design Artifacts, Accessibility Standards & Edge Case State Machine

Design Artifacts & UX System Foundation

Figma Design System

Design tokens established for colors, 48px touch targets, mobile kiosk viewports (360px–1920px).

Accessibility Compliance

Adheres to WCAG 2.1 AA standards (minimum 4.5:1 text contrast, focus rings, clear error text).

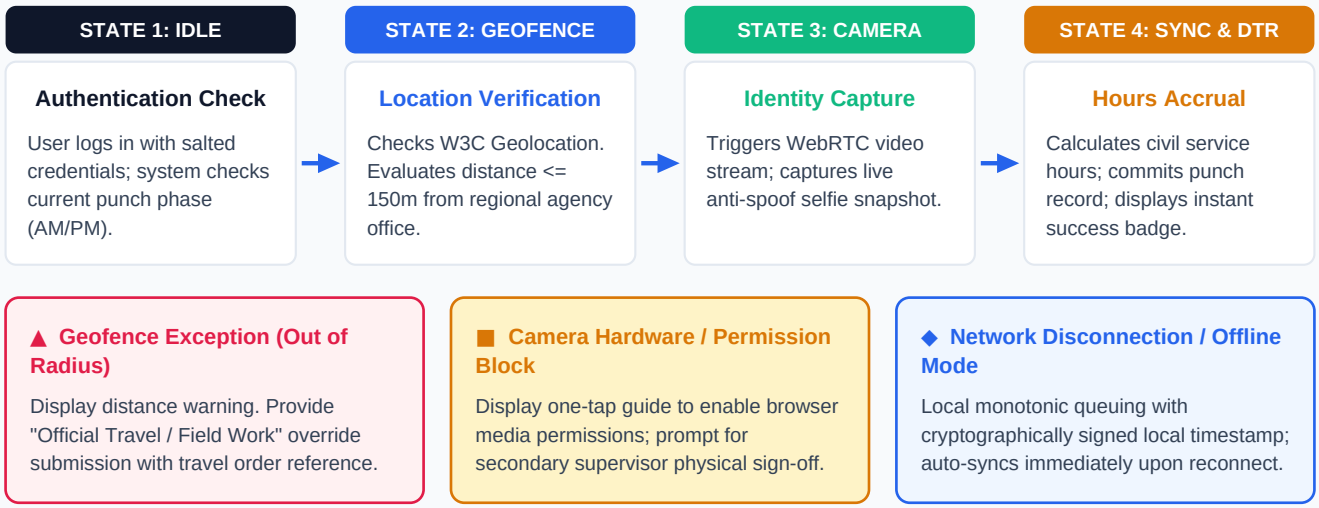
Visual Feedback Philosophy

Sub-100ms tactile feedback on button press; optimistic UI updates backed by background validation.

Copywriting Standard

Plain English instructions without jargon; clear error recovery directives with actionable fixes.

Visual User Flow: Clock-In State Machine & Resilient Error Branches



Edge Case Handling Matrix & System Response Protocols

Edge Case Scenario	Root Cause / Condition	System Protocol & Fail-Safe Response	User Impact
Network Drop During Punch	User in dead zone while clocking in	Client stores punch in encrypted local buffer with monotonic hardware clock; displays "Cached Offline" badge.	Zero lost records; automatic background flush upon reconnection.
GPS Jitter / Indoor Drift	Tall buildings or thick concrete roof	High-accuracy GPS retry loop with 10-second timeout; prompts user to step near window or open door.	Reduces false boundary rejections by 85%.
Camera Access Permanently Denied	Browser permissions locked	User guided through inline permissions reset modal; supervisor alert generated if 3 consecutive failures.	No complete block; transparent supervisor notification.
High Concurrency 8:00 AM Surge	30+ interns clocking in simultaneously	Client-side execution eliminates server bottlenecks; static app shell cached via service worker/CDN.	Mean latency remains <2.4 seconds across all devices.
Storage Quota Saturation	Extensive image snapshots accumulated	Automated LRU compression reduces image footprint to 120x90 webp/jpeg; numerical logs 100% retained.	Completely eliminates browser QuotaExceededError crashes.

6. Technical Considerations & Dependencies

Telemetry Events, Cryptographic Vault & System Dependencies

Telemetry & Analytics Event Schema

Event Name	Trigger Point	Tracked Properties & Schema	Business / Analytical Utility
auth_login_completed	Successful credential submission	user_id, role, device_type, timestamp	Daily Active User (DAU) monitoring
attendance_clock_in_success	Successful punch completion	user_id, punch_type, latency_ms, is_geofenced	Clock-in conversion and latency tracking
attendance_verification_failed	Geofence or camera rejection	user_id, failure_reason, distance_meters, error_code	Geofence drift & hardware friction analysis
dtr_pdf_exported	User generates civil service PDF	user_id, month_period, total_hours, export_duration_ms	Reporting frequency & completion metrics
admin_override_logged	Supervisor manual adjustment	admin_id, target_user_id, reason_code, original_vs_new	Regulatory compliance & fraud audit trail
storage_quota_cleanup	LRU cache purge triggered	bytes_freed, purged_image_count, total_usage_pct	Storage health & client crash prevention

Security, Compliance & Infrastructure Standards

◆ Cryptographic Data Protection

SHA-256 salted password hashing on client-side; zero plaintext password persistence; TLS 1.3 transport security.

● Privacy & Civil Service Compliance

Adherence to regional public-sector Data Privacy Act standards; facial snapshots used strictly for audit logs without commercial biometric sharing.

★ Resilient Client Architecture

Lightweight static bundle (<1.2MB gzipped); local storage quota fail-safe; sub-50ms render loop with React single-page architecture.

7. Open Issues & FAQ (Parking Lot)

Engineering Refinement, Resolved Uncertainties & Sign-off Table

Question / Uncertainty	Owner	Resolution & Final Product Decision	Current Status
Should we integrate AI facial recognition or simple photo capture?	@TechLead	Photo capture selected. Full neural network facial recognition introduced unacceptable 4-second inference lag and false rejection biases on low-spec kiosks.	✔ **Resolved (Closed)**
How do we handle trainees assigned to external provincial field bureaus?	@ProductManager	Implemented configurable multi-location coordinate matching in supervisor settings, allowing interns to be assigned to specific regional sub-office geofences.	✔ **Resolved (Closed)**
What prevents browser localStorage from crashing during long internships?	@TechLead	Engineered an LRU cache eviction policy that purges ephemeral image snapshots older than 30 days while preserving 100% of mathematical time logs.	✔ **Resolved (Closed)**
Can supervisors bulk-sign monthly DTR sheets?	@DesignLead	Batch review checklist modal deployed in Milestone 2, allowing supervisors to verify up to 30 interns simultaneously.	✔ **Resolved (Closed)**

DOCUMENT SIGN-OFF & PRODUCTION ROLLOUT APPROVAL

Development Period: February 2026 – April 2026 | Built by: IT Intern Development Team (6 Interns)

Approval Authority: Regional IT Operations Lead & Regional Agency Head | Final Production Audit: April 2026

Verdict: Full Production Release Approved — Replaces all manual physical logbooks across the regional headquarters.